7 **(a)** State what is meant by total internal reflection.

.....

..... [2]

(b) Fig. 7.1 shows a ray of light from a light source in a tank containing a liquid.

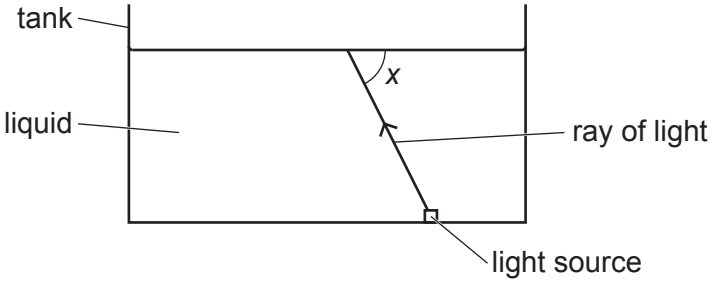


Fig. 7.1

The ray of light strikes the surface of the liquid at an angle *x*.

(i) The refractive index of the liquid is 1.5.

Calculate the largest value of *x* for which total internal reflection can occur.

x = ... [3]

(ii) The speed of light in air is 3.0×10^8 m/s.

Calculate the speed of light in the liquid.

speed = ... [2]

[Total: 7]